

Question 10

Q: An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

Theoretical Concept: Levels of Computer Vision

Sampling defines spatial resolution. If the sampling rate is too low relative to the size of the object, the object will occupy too few pixels to be distinguishable (violating Nyquist).

Image formation factors (like poor focus or motion blur during capture) further spread the object's light across multiple pixels, completely destroying visibility in low-res setups.

OpenCV Python Solution

```
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
print("Upload a road or vehicle image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Create Low-Resolution Image
# -----
small = cv2.resize(img, (80, 80), interpolation=cv2.INTER_AREA)

# Upscale back to original size
low_resolution = cv2.resize(
    small,
```

```

        (img.shape[1], img.shape[0]),
        interpolation=cv2.INTER_NEAREST
    )

# -----
# Sharpen Enhanced Image
# -----
kernel = [[0,-1,0],
          [-1,5,-1],
          [0,-1,0]]

import numpy as np
kernel = np.array(kernel)

enhanced = cv2.filter2D(low_resolution, -1, kernel)

# -----
# Display Images
# -----
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(low_resolution)
plt.title("Low Resolution (Poor Sampling)")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(enhanced)
plt.title("Enhanced Image")
plt.axis("off")

plt.show()

# -----
# Result

```

```
# -----
print("\n===== SAMPLING ANALYSIS =====")
print("Original Resolution :", img.shape[1], "x", img.shape[0])
print("Reduced Sampling      : 80 x 80 pixels")
print("Image Formation       : Reduced object clarity")
print("Enhanced Image         : Improved sharpness")
print("Observation            : Low sampling decreases object visibility.")
print("Conclusion              : Higher sampling improves object detection.")
print("=====")
```

Expected Output

You will see a crisp letter 'A', next to a blurry, unrecognizable gray blob. Because the initial sampling grid (10x10) was too small, the true shape of the object was permanently erased.